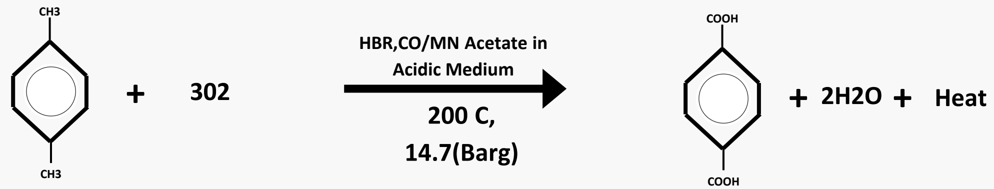
COMPRESSOR-PERFORMANCE-ANALYZER

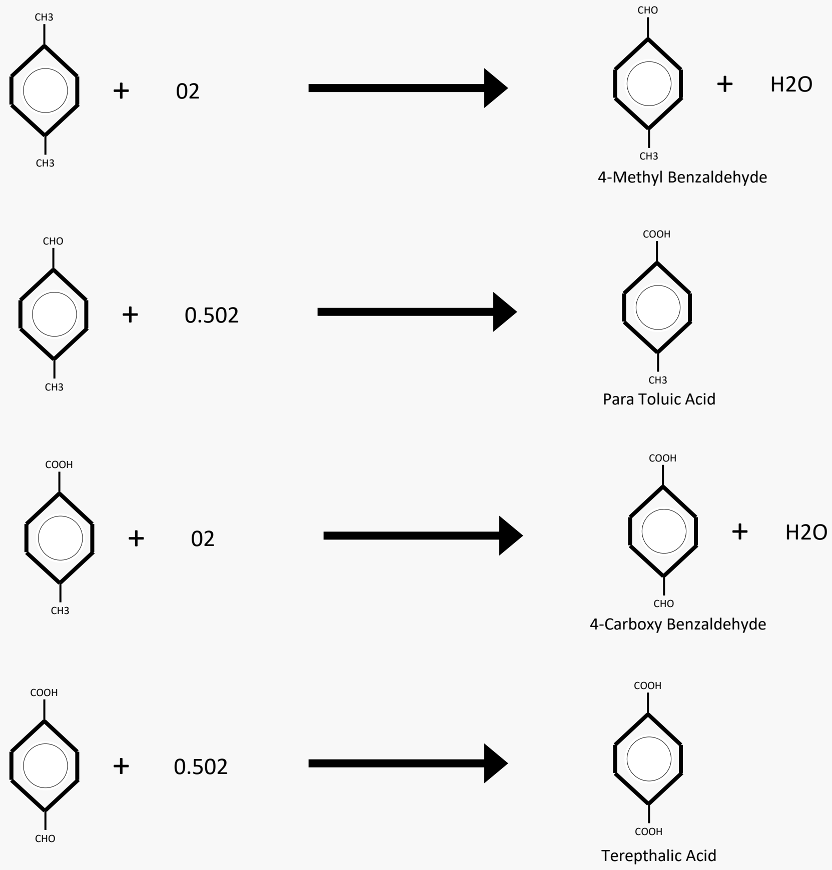
The Compressor Performance Analyzer is a sophisticated, browser-based thermodynamic simulation platform designed to help engineers, researchers, and students accurately analyze, optimize, and forecast the performance of multi-stage centrifugal air compressor systems.

Reaction Pathway for PTA Production From Para-Xylene

Overall Reaction:



Step vise Reaction :



Process Air Compressor (PAC) - Process Description

Configuration of PAC

- Five-stage centrifugal compressor.
- **Intercooler between 1st and 2nd stage.**
- **Intercooler between 2nd and 3rd stage.**
- **Intercooler between 3rd and 4th stage.**
- **Steam Turbine, Process Air Compressor, and Expander are mounted on a common shaft.**
- Inlet Guide Vanes (IGV) provided for flow control.

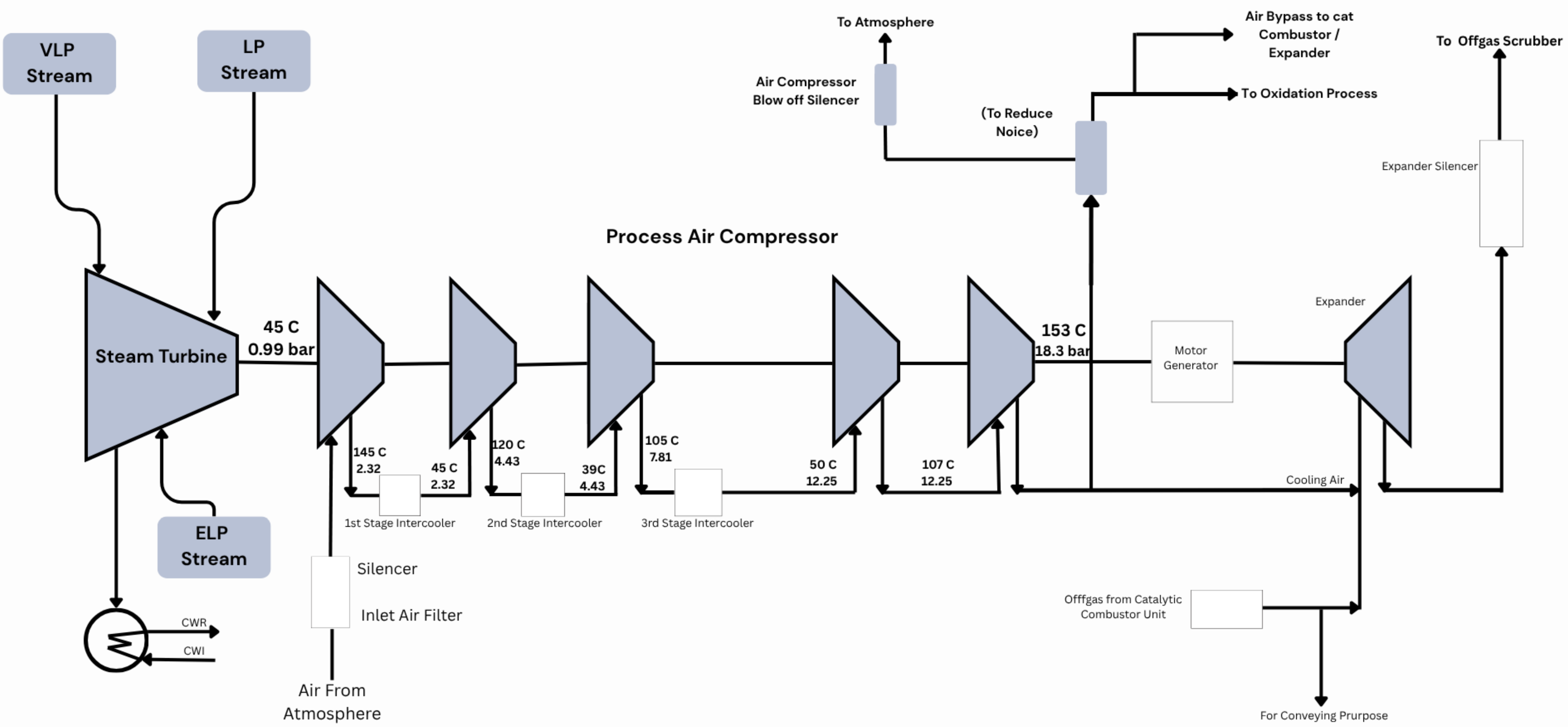
Working Principle

- Atmospheric air enters the compressor through the inlet section.
- **Air is compressed in five stages.**
- The pressure increases progressively in each stage.
- **Compression increases air temperature.**
- Intercoolers remove heat between stages.
- **Compressed air leaves the fifth stage at the required process pressure and temperature.**
- The compressed air is then supplied to downstream process units.

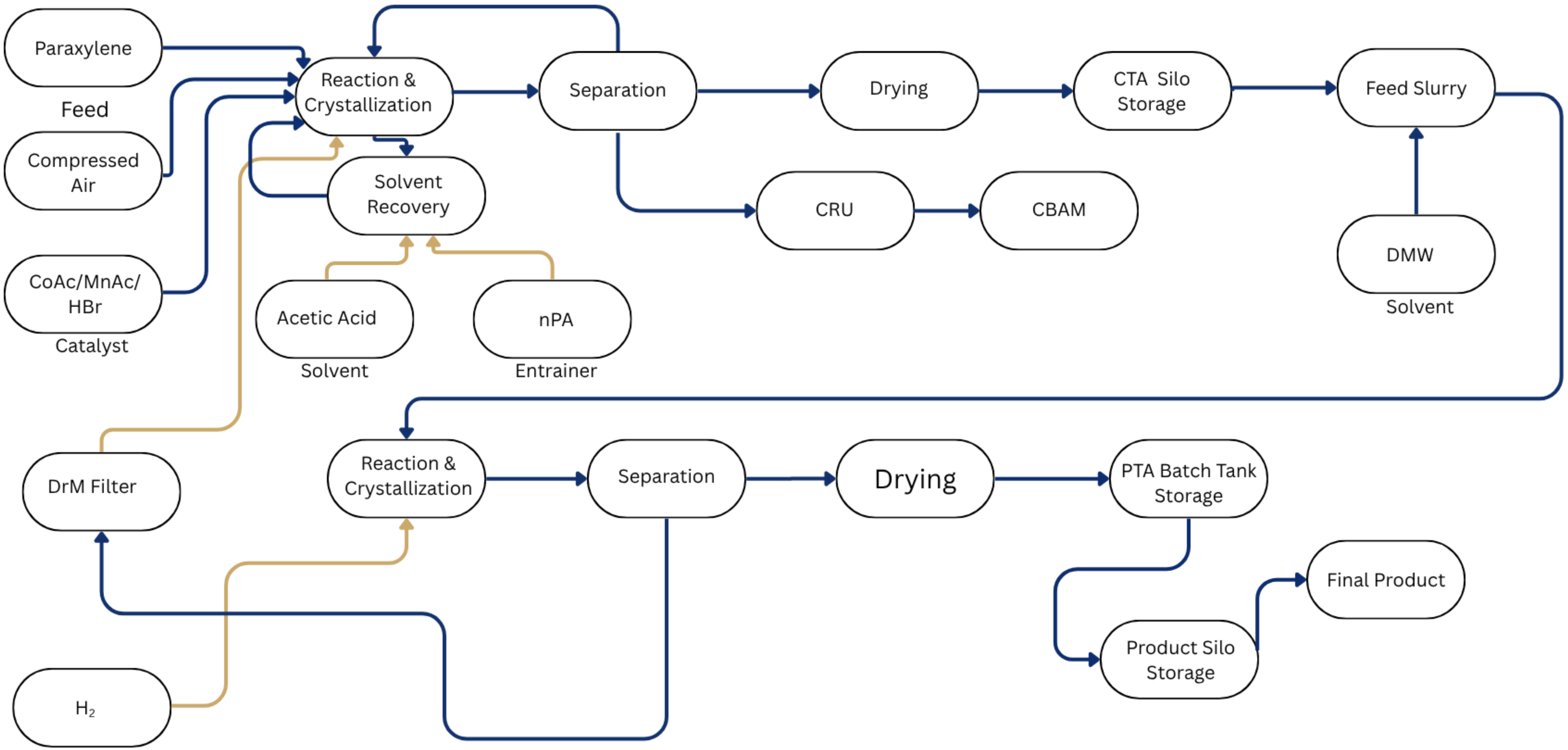
Compressor Drive System

- **During start-up, the compressor is driven by an electric motor.**
- During normal operation, the compressor is driven by a steam turbine and expander.
- **Integrated turbine-expander arrangement enables energy recovery and utilization.**
- This arrangement improves plant energy efficiency.

Process Air Compressor: An Overview



Process Flow Diagram-PTA



Comparison of Rated and Calculated Compressor Performance Data :

Design

Stage	Pin (bar a)	Pout (bar a)	Tin (C)	Tout (C)
1st	0.99	2.319	45	145
2nd	2.319	4.45	40	120
3rd	4.43	8.19	39	105
4th	8.19	12.25	39	107
5th	12.25	18.3	107	153

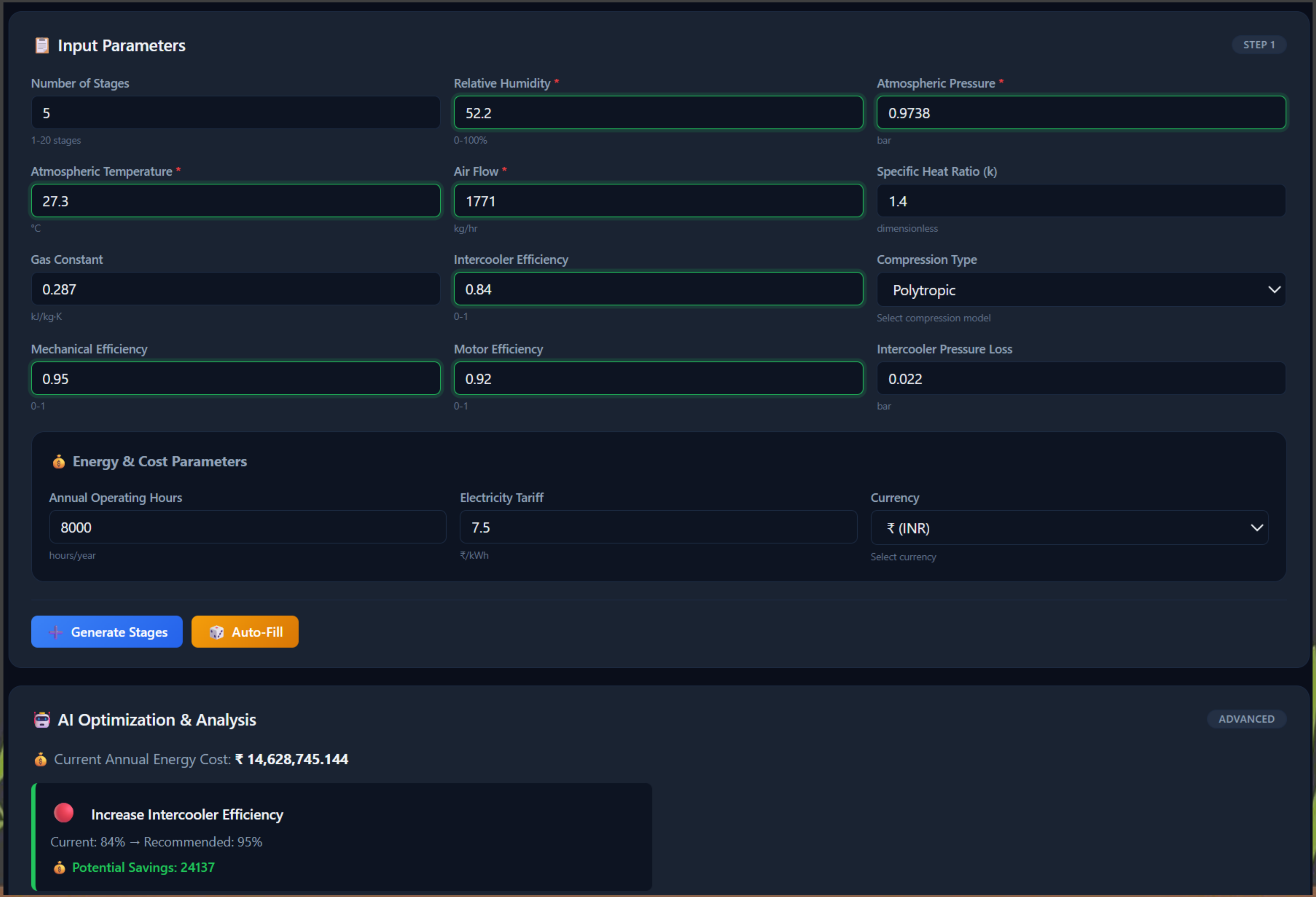
Stage	Kin	Kout	Zin	Zout
1st	1.398	1.3938	.9996	1.0003
2nd	1.4009	1.3978	.9993	1.0012
3rd	1.4059	1.4043	.9988	1.0007
4th	1.4107	1.4099	.9984	1.0012
5th	1.4099	1.4081	1.0012	1.0041

Calculated

Stage	Pin (bar a)	Pout (bar a)	Tin (C)	Tout (C)
1st	0.9932	2.34	34	138.3
2nd	2.3432	4.44	55.8	140.8
3rd	4.4432	7.3532	39.9	107.9
4th	7.3532	12.3932	43.8	106.8
5th	13.05	18.3	106.8	153

Stage	Kin	Kout	Zin	Zout
1st	1.398	1.3938	.9996	1.0003
2nd	1.4009	1.3978	.9993	1.0012
3rd	1.4059	1.4043	.9988	1.0007
4th	1.4107	1.4099	.9984	1.0012
5th	1.4099	1.4081	1.0012	1.0041

App Screenshot



Comparison of Design and Calculated Inlet Air Conditions :

Design

Relative Humidity	60	%
Atomspheric Pressure	0.99	bar
Suction Pressure	**99000**	Pa
Temperaturein	**45**	C
Vapour Pressure of Water Vapour	**9552**	Pa
Partial Pressure of Water Vapour	**5731**	Pa
Mole Fraction(Xw) of water in	**0.05789**	

Calculated

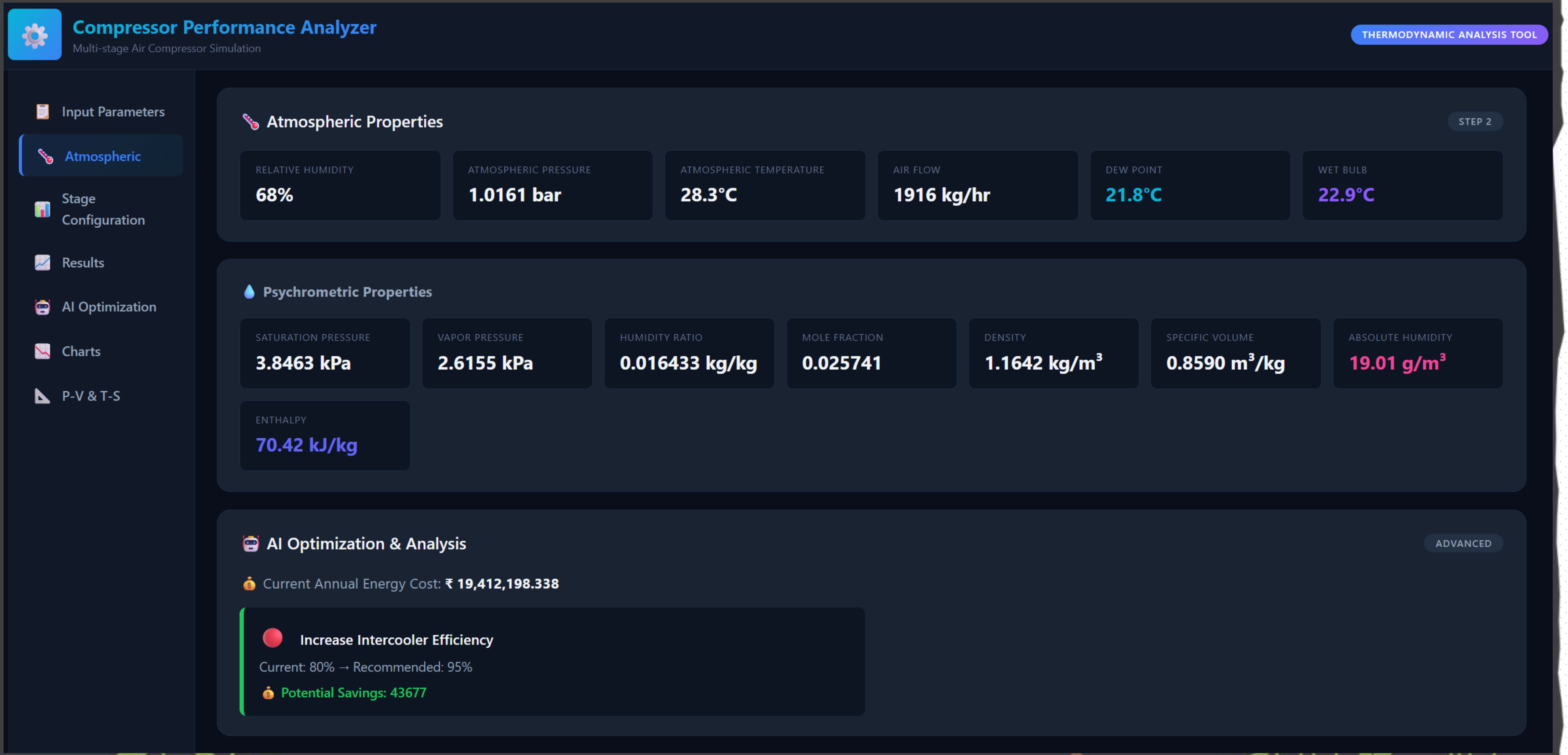
Relative Humidity	65	%
Atomspheric Pressure	0.9932	bar
Suction Pressure	**99320**	Pa
Temperaturein	**34**	C
Vapour Pressure of Water Vapour	**5302**	Pa
Partial Pressure of Water Vapour	**3446**	Pa
Mole Fraction(Xw) of water in	**0.03470**	

Stage-wise Water Vapour Mole Fraction Analysis (Design Case) :

Design

Stage	Xwin	Pin	Tin	Pstageout	Tstageout	Pcoolerstage,out	Tcoolerstage,out	V.Pstage out	Xwout	Remarks
1	0.05789	0.99	45	2.32	145	2.319	40	7352.789	0.03171	Condensation
2	0.03171	2.319	40	4.43	120	4.43	39	6969.77	.01573	Condensation
3	0.01573	4.43	39	8.19	105	7.81	50	12296.2	.01501	Condensation
4	0.01501	8.19	50	12.25	107	12.25	107	129570.3	.01501	No Condensation
5	0.01501	13.05	107	18.3	153	18.3	153	511403	.01501	No Condensation

App Screenshot



Stage-wise Water Vapour Mole Fraction Analysis (Calculated Case) :

Calculated

Stage	**Xw,in**	P,in	T,in	Pstageout	Tstageout	Pcoolerst age,out	Tcoolerst age,out	**VP stage out**	**Xw,out**	Remarks
1	0.03470	0.9932	34	2.34	138.3	2.3432	55.8	16301.2	0.03470	**No Condensation**
2	0.03470	2.3432	55.8	4.44	140.8	4.4432	39.9	7313.687	0.01646	Condensation
3	0.01646	4.4432	39.9	7.3532	107.9	7.81	43.8	8978.649	0.01221	Condensation
4	0.01221	7.3532	43.8	12.3932	106.8	12.3932	106.8	128698.1	0.01221	No Condensation
5	0.01221	13.05	106.8	18.3	153	18.3	153	511403.4	.01221	No Condensation

App Screenshot

Simulation Results

STEP 4

Export | CSV | PDF | Compare | Clear

TOTAL POWER	SHAFT POWER	MOTOR POWER	OVERALL EFFICIENCY	TOTAL CONDENSATION	FREE AIR DELIVERY	TOTAL HEAD
243.81 kW	256.64 kW	278.96 kW	86.5%	0.78 kg/hr	1445.45 m^3/hr	495.71 kJ/kg

FINAL AIR FLOW
1770.22 kg/hr

ANNUAL OPERATING HOURS	ELECTRICITY TARIFF	ANNUAL ENERGY CONSUMPTION	ANNUAL ENERGY COST
8,000 hrs	₹ 7.50/kWh	1,950,499.353 kWh	₹ 14,628,745.144

Carbon Footprint Analysis

ANNUAL CO_2 EMISSIONS	EMISSION FACTOR	EFFICIENCY RATING	TREES NEEDED TO OFFSET
1599.4 tons	0.82 kg CO_2/kWh	POOR	76162 trees

Industry Benchmark: Excellent: <50 tons/yr | Good: 50-100 tons/yr | Average: 100-150 tons/yr | Poor: >150 tons/yr

ANNUAL OPERATING HOURS	ELECTRICITY TARIFF	ANNUAL ENERGY CONSUMPTION	ANNUAL ENERGY COST
8,000 hrs	₹ 7.50/kWh	1,950,499.353 kWh	₹ 14,628,745.144

STAGE	PR	TR	N	H_POLY (%)	H_VOL (%)	POWER (KW)	HEAD (KJ/KG)	COND (KG/HR)	MOIST (KG/HR)
1	2.466	1.131	2.500	92.0	95.5	52.03	105.77	0.00	1771.00
2	2.286	1.076	2.500	92.0	96.0	45.95	93.41	0.39	1770.61
3	2.345	1.068	2.500	92.0	95.8	46.94	95.43	0.30	1770.32
4	2.533	1.110	2.500	92.0	95.3	53.85	109.51	0.00	1770.32
5	2.272	1.061	2.500	92.0	96.0	45.04	91.59	0.10	1770.22

Design Case - Stage-wise Water Condensation :

Design

Stage	**Xw,in**	**Xw,out**	Remarks	Water In (kmol/hr)	Water Out (kmol/hr)	Water Condensed (kg/hr)
1	0.05789	0.03171	**Condensation**	1039.725	554.026	8742.6
2	0.03171	.01573	**Condensation**	554.026	270.451	5104.4
3	0.01573	.01501	**Condensation**	270.451	257.895	226.0
4	0.01501	.01501	No Condensation	257.895	257.895	0.0
5	0.01501	.01501	No Condensation	257.895	257.895	0.0

Comparison of Mass of Water in Air at Suction Condition (Design vs. Calculated) :

Design

Mw of water from 5th stage discharge	28.80264542	g/mol	--	--
Moles of Wet Air	17959.16229	Kmole/hr	510.0402	t/hr
moles of dry air	16919.43686	kmole/hr	480.512	t/hr
Mass of water in air at suction condition	---	---	29.5282	t/hr

Calculated

Mw of water from 5th stage discharge	28.8333313	g/mol	--	--
Moles of Wet Air	18662.46449	Kmole/hr	530.014	t/hr
moles of dry air	18014.88983	kmole/hr	511.6229	t/hr
Mass of water in air at suction condition	---	---	18.39113	t/hr

Calculated Case - Stage-wise Water Condensation :

Calculated

Stage	**Xw,in**	**Xw,out**	Remarks	**Water In (kmol/hr)**	**Water Out (kmol/hr)**	**Water Condensed (kg/hr)**
1	0.03470	0.03470	**No Condensation**	647.575	647.575	0.0
2	0.03470	0.01646	Condensation	647.575	301.495	6229.4
3	0.01646	0.01221	Condensation	301.495	222.691	1418.5
4	0.01221	0.01221	No Condensation	222.691	222.691	0.0
5	0.01221	.01221	No Condensation	222.691	222.691	0.0

Comparison of Stage-wise Water Condensation Across Intercoolers (Design vs. Calculated) :

Design

Result Of all 5 stages :

Condensation in 1st stage(E-116)	8.75	t/hr
Condensation in 2nd stage(E-117)	5.11	t/hr
Condensation in 3rd stage(E-118)	0.23	t/hr
Total Water Condensed	14.0886	t/hr
Remaining Water in air	15.4	t/hr
Total Remaining air Flow	496.0	t/hr

Calculated

Result Of all 5 stages :

Condensation in 1st stage(E-116)	0.00	t/hr
Condensation in 2nd stage(E-117)	6.24	t/hr
Condensation in 3rd stage(E-118)	1.42	t/hr
Total Water Condensed	7.6564	t/hr
Remaining Water in air	10.7	t/hr
Total Remaining air Flow	522.4	t/hr

Stage-wise Compressor Efficiency Analysis (Design Case) :

Design

Result Of all 5 stages :

Stage	Efficiency(%)	Kavg.	Pout/Pin	(K-1)/K	Tout/Tin	Zout/Zin
1	**88.1**	1.396	2.342	0.284	1.314	0.99685
2	**80.5**	1.39935	1.910	0.285	1.256	1.00190
3	**84.4**	1.4051	1.763	0.288	1.212	1.00190
4	**82.0**	1.4103	1.593	0.291	1.176	1.00280
5	**83.8**	1.4090	1.402	0.290	1.121	1.00290

Stage-wise Compressor Efficiency Analysis (Calculated Case) :

Calculated

Result Of all 5 stages :

Stage	Kavg.	Pout/Pin	(K-1)/K	Tout/Tin	Zout/Zin	Efficiency(%)
1	1.396	2.359	0.284	1.340	0.99685	**84.2**
2	1.39935	1.896	0.285	1.256	1.00190	**78.8**
3	1.4051	1.758	0.288	1.217	1.00190	**81.9**
4	1.4103	1.685	0.291	1.199	1.00280	**82.5**
5	1.4090	1.402	0.290	1.22	1.00290	**83.4**

App Screenshot

App Screenshot

Stage-wise Polytropic and Actual Power Analysis (Design Case) :

Design

Result Of all 5 stages :

Stage	Actual Power(MW)	Polytopic Power(KW)	Polytropic exponent, n	n/(n-1)	Mw inlet	Zin	Tin(K)	AirFlow(Kg/s)
1	**14.43**	**12891**	**1.4665**	3.149	28.33	0.9996	318.15	**141.34**
2	**11.42**	**9188**	**1.549**	2.820	28.62	0.9993	313.15	**138.91**
3	**9.18**	**7744**	**1.519**	2.926	28.79	0.9988	312.15	**137.49**
4	**7.91**	**6484**	**1.550**	2.818	28.80	0.9984	323.15	**137.43**
5	**6.47**	**5416**	**1.530**	2.886	28.80	1.0012	380.15	**137.43**

Stage-wise Polytropic and Actual Power Analysis (Calculated Case) :

Calculated

Result Of all 5 stages :

Stage	Polytropic exponent, n	n/(n-1)	Mw inlet	Zin	Tin(K)	AirFlow(Kg/s)	Polytopic Power(KW)	Actual Power(MW)
1	1.508	2.967	28.59	0.9996	307.15	148.20	13172	15.65
2	1.568	2.760	28.59	0.9993	328.95	148.20	10202	12.95
3	1.543	2.841	28.79	0.9988	313.05	146.46	8252	10.08
4	1.545	2.834	28.83	0.9984	316.95	146.07	7639	9.26
5	1.534	2.873	28.83	1.0012	379.95	146.07	5749	6.89

Findings :

Stage	Efficiency Observation	Result	Conclusion
1	Design: **88.1%** Actual: **84.2%**	**3.9%** efficiency reduction.	Highest performance deterioration; priority for improvement.
2	Design: **80.5%** Actual: **78.8%**	**1.7%** efficiency reduction.	Indicates higher thermodynamic losses and requires performance optimization.
3	Design: **84.4%** Actual: **81.9%**	**2.5%** efficiency reduction.	Moderate performance deterioration.
4	Design: **82.0%** Actual: **82.5%**	**0.5%** efficiency improvement	Operating close to design conditions.
5	Design: **83.8%** Actual: **83.4%**	**0.4%** efficiency reduction.	Stable compressor performance.
Overall PAC	Design: **83.76%**	**1.6%** overall efficiency	Performance degradation is mainly concentrated in

Result of all the table :

Design

Result Of all 5 stages :

Stage	Air Flow(kg/s)	Actual power (MW)	Efficiency	Water condensation (Kg/hr)	Polytropic power(KW)
1	141.34	14.67	88.1	8742.6	12917
2	138.91	11.42	80.5	5104.4	9188
3	137.49	9.18	84.4	226.00	7744
4	137.43	7.91	82.0	0.0	6484
5	137.43	6.47	83.8	0.0	5416

Calculated

Result Of all 5 stages :

Stage	Water condensation	Air Flow(kg/s)	Polytropic power(KW)	Actual power (MW)	Efficiency
1	0.0	148.20	13172	15.65	84.2
2	6229.4	148.20	10202	12.95	78.8
3	1418.5	146.46	8252	10.08	81.9
4	0.0	146.07	7639	9.26	82.5
5	0.0	146.07	5749	6.89	83.4

Result of all stage water condensation :

Design

Result Of all 5 stages :

Water condensed in 1st stage	8742.6	kg/hr
Water condensed in 2nd stage	5104.4	kg/hr
Water condensed in 3rd stage	226.00	kg/hr
Water condensed in 4th stage	0.0	kg/hr
Water condensed in 5th stage	0.0	kg/hr

Calculated

Result Of all 5 stages :

Water condensed in 1st stage	0.0	kg/hr
Water condensed in 2nd stage	6229.4	kg/hr
Water condensed in 3rd stage	1418.5	kg/hr
Water condensed in 4th stage	0.0	kg/hr
Water condensed in 5th stage	0.0	kg/hr

Future Plans

- **Shell-Side and Tube-Side Heat Balance Evaluation of intercoller.**
- Determination of the Polytropic Exponent of the PAC Compressor in PTA-5/6.
- **Polytropic Power Calculation of the PAC Compressor in PTA-5/6.**
- **Actual Power Consumption Analysis of the PAC Compressor in PTA-5/6.**
- Find which factors affecting PAC compressor efficiency and how we can resolve it.
- **What are the major causes of efficiency losses in the PAC system.**
- Impact of suction conditions, cooling efficiency, and pressure drops on overall efficiency as well as Weather.
- **Recommendations for improving PAC compressor performance and operational reliability.**

Recommendation

- **Ensure proper installation, alignment**, and **handling** of **air seal strips during maintenance** and **shutdown** activities to prevent damage and **air bypassing.**
- Carry out **periodic cleaning and inspection** of **finned tube bundles** to maintain effective heat transfer performance.
- Monitor **intercooler outlet temperature**, **pressure drop**, and **condensate** generation regularly to detect performance deterioration at an early stage.
- Maintain adequate **cooling water flow** and quality to **minimize fouling** and sustain intercooler effectiveness.
- Conduct periodic **compressor performance audits** to identify **efficiency losses** and implement corrective actions proactively.
- Adopt a **condition-based maintenance approach for intercoolers** and **compressor internals** to prevent unexpected **failures** and **efficiency degradation.**